

Mark Scheme

Q1.

Part	Working or answer an examiner might expect to see	Mark	Notes
(a)	$V = Ae^{-kt}$	M1	This mark is given for suggesting a suitable exponential model for V in terms of t
	When $t = 0$ and $V = 20\,000$, $A = 20\,000$	M1	This mark is given for using the model to show the initial value for A is £20 000
	When $t = 1$ and $V = 16\,000$, $16\,000 = 20\,000e^{-1k}$ $k = \ln 0.8 = -0.223$	M1	This mark is given for using the value of the car after one year to find a value for k
	$V = 20\,000e^{-0.223t}$	A1	This mark is given for finding a fully correct exponential model
(b)	When $t = 10$, $V = £2150$	M1	This mark is given for finding a value for V when $t = 10$
	This model is reliable since the value £2150 is close to £2000	A1	This mark is given for a valid statement comparing the two possible values of the car after 10 years
(c)	For example: The value of k should be increased (e.g. $V = 20\,000e^{-0.1t}$) A constant should be added (e.g. $V = 20\,000e^{-0.223t} + 2000$)	B1	This mark is given for a statement suggesting a valid adaptation

Q2.

Question	Scheme	Marks	AOs
(a)	$A = 1000$	B1	3.4
	$2000 = 1000e^{5k}$ or $e^{5k} = 2$	M1	1.1b
	$e^{5k} = 2 \Rightarrow 5k = \ln 2 \Rightarrow k = \dots$	M1	2.1
	$N = 1000e^{\left(\frac{1}{5}\ln 2\right)t}$ or $N = 1000e^{0.139t}$	A1	3.3
		(4)	
(b)	$\frac{dN}{dt} = 1000 \times \left(\frac{1}{5}\ln 2\right)e^{\left(\frac{1}{5}\ln 2\right)t}$ or $\frac{dN}{dt} = 1000 \times 0.139e^{0.139t}$ $\left(\frac{dN}{dt}\right)_{t=8} = 1000 \times \left(\frac{1}{5}\ln 2\right)e^{8 \times \frac{1}{5}\ln 2}$ or $\left(\frac{dN}{dt}\right)_{t=8} = 1000 \times 0.139e^{0.139 \times 8}$	M1	3.1b
	= awrt 420	A1	1.1b
		(2)	
(c)	$500e^{1.4 \times \left(\frac{1}{5}\ln 2\right)T} = 1000e^{\left(\frac{1}{5}\ln 2\right)T}$ or $500e^{1.4 \times 0.139T} = 1000e^{0.139T}$	M1	3.4
	Correct method of getting a linear equation in T E.g. $0.08T \ln 2 = \ln 2$ or $1.4 \times 0.139T = \ln 2 + 0.139T$	M1	2.1
	$T = 12.5$ hours	A1	1.1b
		(3)	
(9 marks)			
Notes			

Mark as one complete question. Marks in (a) can be awarded from (b)

(a)

B1: Correct value of A for the model. Award if equation for model is of the form $N = 1000e^{-t}$

M1: Uses the model to set up a correct equation in k . Award for substituting $N = 2000, t = 5$

following through on their value for A .

M1: Uses correct $\ln$ work to solve an equation of the form $ae^{5k} = b$ and obtain a value for k

A1: Correct equation of model. Condone an ambiguous $N = 1000e^{\frac{1}{5}\ln 2t}$ unless followed by something incorrect. Watch for $N = 1000 \times 2^{\frac{1}{5}t}$ which is also correct

(b)

M1: Differentiates ae^{kt} to βe^{kt} and substitutes $t = 8$ (Condone $\alpha = \beta$ so long as you can see an attempt to differentiate)

A1: For awrt 420 (2sf).

(c)

M1: Uses both models to set up an equation in T using their value for k , but also allow in terms of k

M1: Uses correct processing using $\ln$ s to obtain a linear equation in T (or t)

A1: Awrt 12.5

Answers to (b) and (c) appearing without working (i.e. from a calculator).

It is important that candidates show sufficient working to make their methods clear.

(b) If candidate has for example $N = 1000e^{0.139t}$, and then writes at $t = 8$ $\frac{dN}{dt} = \text{awrt } 420$ award both marks. Just the answer from a correct model equation score SC 1,0.

(c) The first M1 should be seen E.g. $500e^{1.4 \times 0.139T} = 1000e^{0.139T}$

If the answer $T = 12.5$ appears without any further working score SC M1 M1 A0

Q3.

Question	Scheme	Marks	AOs
	$N = \frac{900}{3 + 7e^{-0.25t}} = 900(3 + 7e^{-0.25t})^{-1}, t \in \mathbb{R}, t \geq 0; \frac{dN}{dt} = \frac{N(300 - N)}{1200}$		
(a)	90	B1	3.4
		(1)	
(b) Way 1	$\frac{dN}{dt} = -900(3 + 7e^{-0.25t})^{-2} (7(-0.25)e^{-0.25t}) \left\{ = \frac{900(0.25)(7)e^{-0.25t}}{(3 + 7e^{-0.25t})^2} \right\}$	M1	2.1
		A1	1.1b
	$\Rightarrow \frac{dN}{dt} = \frac{900(0.25) \left(\left(\frac{900}{N} - 3 \right) \right)}{\left(\frac{900}{N} \right)^2}$	dM1	2.1
	correct algebra leading to $\frac{dN}{dt} = \frac{N(300 - N)}{1200}$ *	A1*	1.1b
		(4)	
(b) Way 2	$\frac{dN}{dt} = -900(3 + 7e^{-0.25t})^{-2} (7(-0.25)e^{-0.25t}) \left\{ = \frac{900(0.25)(7)e^{-0.25t}}{(3 + 7e^{-0.25t})^2} \right\}$	M1	2.1
		A1	1.1b
	$\frac{N(300 - N)}{1200} = \frac{\left(\frac{900}{3 + 7e^{-0.25t}} \right) \left(300 - \frac{900}{3 + 7e^{-0.25t}} \right)}{1200}$	dM1	2.1
	LHS = $\frac{1575e^{-0.25t}}{(3 + 7e^{-0.25t})^2}$ o.e., RHS = $\frac{900(300(3 + 7e^{-0.25t}) - 900)}{1200(3 + 7e^{-0.25t})^2} = \frac{1575e^{-0.25t}}{(3 + 7e^{-0.25t})^2}$ o.e. and states hence $\frac{dN}{dt} = \frac{N(300 - N)}{1200}$ (or LHS = RHS) *	A1*	1.1b
		(4)	
(c)	Deduces $N = 150$ (can be implied)	B1	2.2a
	so $150 = \frac{900}{3 + 7e^{-0.25T}} \Rightarrow e^{-0.25T} = \frac{3}{7}$	M1	3.4
	$T = -4 \ln\left(\frac{3}{7}\right)$ or $T = \text{awrt } 3.4$ (months)	dM1	1.1b
		A1	1.1b
		(4)	
(d)	either one of 299 or 300	B1	3.4
		(1)	
(10 marks)			

Notes for Question	
(b)	
M1:	Attempts to differentiate using - the chain rule to give $\frac{dN}{dt} = \pm Ae^{-0.25t}(3+7e^{-0.25t})^{-2}$ or $\frac{\pm Ae^{-0.25t}}{(3+7e^{-0.25t})^2}$ o.e. - the quotient rule to give $\frac{dN}{dt} = \frac{(3+7e^{-0.25t})(0) \pm Ae^{-0.25t}}{(3+7e^{-0.25t})^2}$ - implicit differentiation to give $N(3+7e^{-0.25t}) = 900 \Rightarrow (3+7e^{-0.25t})\frac{dN}{dt} \pm ANe^{-0.25t} = 0$, o.e. where $A \neq 0$
Note:	Condone a slip in copying $(3+7e^{-0.25t})$ for the M mark
A1:	A correct differentiation statement
Note:	Implicit differentiation gives $(3+7e^{-0.25t})\frac{dN}{dt} - 1.75Ne^{-0.25t} = 0$
dM1:	Way 1: Complete attempt, by eliminating t , to form an equation linking $\frac{dN}{dt}$ and N only Way 2: Complete substitution of $N = \frac{900}{3+7e^{-0.25t}}$ into $\frac{dN}{dt} = \frac{N(300-N)}{1200}$
Note:	Way 1: e.g. substitutes $3+7e^{-0.25t} = \frac{900}{N}$ and $e^{-0.25t} = \frac{900}{N} - 3$ or substitutes $e^{-0.25t} = \frac{\frac{900}{N} - 3}{7}$ into their $\frac{dN}{dt} = \dots$ to form an equation linking $\frac{dN}{dt}$ and N
A1*:	Way 1: Correct algebra leading to $\frac{dN}{dt} = \frac{N(300-N)}{1200}$ * Way 2: See scheme
(c)	
B1:	Deduces or shows that $\frac{dN}{dt}$ is maximised when $N = 150$
M1:	Uses the model $N = \frac{900}{3+7e^{-0.25t}}$ with their $N = 150$ and proceeds as far as $e^{-0.25t} = k, k > 0$ or $e^{0.25t} = k, k > 0$. Condone $t \equiv T$
dM1:	Correct method of using logarithms to find a value for T . Condone $t \equiv T$
A1:	see scheme
Note:	$\frac{d^2N}{dt^2} = \frac{dN}{dt} \left(\frac{300}{1200} - \frac{2N}{1200} \right) = 0 \Rightarrow N = 150$ is acceptable for B1
Note:	Ignore units for T
Note:	Applying $300 = \frac{900}{3+7e^{-0.25t}} \Rightarrow t = \dots$ or $0 = \frac{900}{3+7e^{-0.25t}} \Rightarrow t = \dots$ is M0 dM0 A0
Note:	M1 dM1 can only be gained in (c) by using an N value in the range $90 < N < 300$
(d)	
B1:	300 (or accept 299)



Question	Scheme	Marks	AOs
	$N = \frac{900}{3 + 7e^{-0.25t}} = 900(3 + 7e^{-0.25t})^{-1}, t \in \mathbb{R}, t \geq 0; \quad \frac{dN}{dt} = \frac{N(300 - N)}{1200}$		
(b) Way 3	$\int \frac{1}{N(300 - N)} dN = \int \frac{1}{1200} dt$	M1	2.1
	$\int \frac{1}{300} \left(\frac{1}{N} + \frac{1}{300 - N} \right) dN = \int \frac{1}{1200} dt$	A1	1.1b
	$\frac{1}{300} \ln N - \frac{1}{300} \ln(300 - N) = \frac{1}{1200} t \{+c\}$		
	$\{t = 0, N = 90 \Rightarrow\} \quad c = \frac{1}{300} \ln(90) - \frac{1}{300} \ln(210) \Rightarrow c = \frac{1}{300} \ln\left(\frac{3}{7}\right)$ $\frac{1}{300} \ln N - \frac{1}{300} \ln(300 - N) = \frac{1}{1200} t + \frac{1}{300} \ln\left(\frac{3}{7}\right)$ $\ln N - \ln(300 - N) = \frac{1}{4} t + \ln\left(\frac{3}{7}\right)$ $\ln\left(\frac{N}{300 - N}\right) = \frac{1}{4} t + \ln\left(\frac{3}{7}\right) \Rightarrow \frac{N}{300 - N} = \frac{3}{7} e^{\frac{1}{4}t}$	dM1	2.1
	$7N = 3e^{\frac{1}{4}t}(300 - N) \Rightarrow 7N + 3Ne^{\frac{1}{4}t} = 900e^{\frac{1}{4}t}$ $N(7 + 3e^{\frac{1}{4}t}) = 900e^{\frac{1}{4}t} \Rightarrow N = \frac{900e^{\frac{1}{4}t}}{7 + 3e^{\frac{1}{4}t}} \Rightarrow N = \frac{900}{3 + 7e^{-0.25t}} *$	A1*	1.1b
		(4)	
(b) Way 4	$N(3 + 7e^{-0.25t}) = 900 \Rightarrow e^{-0.25t} = \frac{1}{7} \left(\frac{900}{N} - 3 \right) \Rightarrow e^{-0.25t} = \frac{900 - 3N}{7N}$	M1	2.1
	$\Rightarrow t = -4(\ln(900 - 3N) - \ln(7N))$ $\Rightarrow \frac{dt}{dN} = -4 \left(\frac{-3}{900 - 3N} - \frac{7}{7N} \right)$	A1	1.1b
	$\frac{dt}{dN} = 4 \left(\frac{1}{300 - N} + \frac{1}{N} \right) \Rightarrow \frac{dt}{dN} = 4 \left(\frac{N + 300 - N}{N(300 - N)} \right)$	dM1	2.1
	$\frac{dt}{dN} = \left(\frac{1200}{N(300 - N)} \right) \Rightarrow \frac{dN}{dt} = \frac{N(300 - N)}{1200} *$	A1*	1.1b
		(4)	

Notes for Question Continued

(b) Way 3	
M1:	Separates the variables, an attempt to form and apply partial fractions and integrates to give $\ln$ terms = $kt \{+c\}$, $k \neq 0$, with or without a constant of integration c
A1:	$\frac{1}{300} \ln N - \frac{1}{300} \ln(300 - N) = \frac{1}{1200} t \{+c\}$ or equivalent with or without a constant of integration c
dM1:	Uses $t = 0, N = 90$ to find their constant of integration and obtains an expression of the form $\lambda e^{\frac{1}{4}t} = f(N)$; $\lambda \neq 0$ or $\lambda e^{-\frac{1}{4}t} = f(N)$; $\lambda \neq 0$
A1*:	Correct manipulation leading to $N = \frac{900}{3 + 7e^{-0.25t}} *$
(b) Way 4	
M1:	Valid attempt to make t the subject, followed by an attempt to find two $\ln$ derivatives, condoning sign errors and constant errors.
A1:	$\frac{dt}{dN} = -4 \left(\frac{-3}{900 - 3N} - \frac{7}{7N} \right)$ or equivalent
dM1:	Forms a common denominator to combine their fractions
A1*:	Correct algebra leading to $\frac{dN}{dt} = \frac{N(300 - N)}{1200} *$

Q4.

Question	Scheme	Marks	AOs
(a)	(i) Method to find p Eg. Divides $32000 = Ap^4$ by $50000 = Ap^{11}$ $p^7 = \frac{50000}{32000} \Rightarrow p = \sqrt[7]{\frac{50000}{32000}} = \dots$	M1	3.1a
	$p = 1.0658$	A1	1.1b
	(ii) Substitutes their $p = 1.0658$ into either equation and finds A $A = \frac{32000}{1.0658^4} \text{ or } A = \frac{50000}{1.0658^{11}}$	M1	1.1b
	$A = 24795 \rightarrow 24805 \approx 24\,800^*$	A1*	1.1b
		(4)	
(b)	A / (£) 24 800 is the value of the car on 1st January 2001	B1	3.4
	$p/1.0658$ is the factor by which the value rises each year. Accept that the value rises by 6.6 % a year (ft on their p)	B1	3.4
		(2)	
(c)	Attempts $100000 = 24800 \times 1.0658^t$		
	$1.0658^t = \frac{100000}{24800}$	M1	3.4
	$t = \log_{1.0658} \left(\frac{100000}{24800} \right)$	dM1	1.1b
	$t = 21.8 \text{ or } 21.9$	A1	1.1b
	cs0 2022	A1	3.2a
		(4)	
(10 marks)			

(a) (i) M1: Attempts to use both pieces of information within $V = Ap^t$, eliminates A correctly and solves an equation of the form $p^n = k$ to reach a value for p. Allow for slips on the 32 000 and 50 000 and the values of t. A1: $p = \text{awrt } 1.0658$ Both marks can be awarded from incorrect but consistent interpretations of t. Eg. $32000 = Ap^5$, $50000 = Ap^{12}$ (a)(ii) M1: Substitutes their $p = 1.0658$ into either of their equations and finds A Eg $A = \frac{32000}{1.0658^4}$ or $A = \frac{50000}{1.0658^7}$ but you may follow through on incorrect equations from part (i) A1*: Shows that A is between 24 795 and 24 805 before you see '$\approx 24\,800$' or '≈ 24800'. Accept with or without units. An alternative to (ii) is to start with the given answer. M1: Attempts $24800 \times 1.0658^4 = (32000.34)$
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A1: 24800×1.0658^4 , achieves a value between 31095 and 32005 followed by $\approx 32\,000$ hence A must be $\approx 24\,800$

(b)

B1: States that A is the value of the car on 1st January 2001.
 The statement must reference **the car**, its **cost/value**, and **"0" time**
 Allow 'it is the initial value of the car' 'it is the cost of the car at $t = 0$ ' 'it is the cars starting value'

B1: States that p is the rate at which the value of the car rises each year.
 The statement must reference **a yearly rate** and **an increase in value or multiplier**.
 They could reference the 1.0658 Eg "The cars value rises by 6.5 % each year."
 Allow " p is the rate the cars value is rising each year" "it is the proportional increase in value of the car each year" "the factor by which the value of the car is rising each year" 'its value appreciates by 6.5% per year' Allow 'the value of the car multiplies by p each year'
 Do not allow "by how much the value of the car rises each year" or "it is the rate of inflation"

(c)

M1: Uses the model $100000 = 24800 \times 1.0658^t$ and proceeds to their $1.0658^t = k$
 Allow use of any inequality here.

dM1: For the complete method of (i) using the information given with their equation of the model and (ii) translating the situation into a correct method to find ' t '

A1: $(t) = \text{awrt } 21.8 \text{ or } 21.9 \text{ or } \log_{1.0658} \left(\frac{100000}{24800} \right) \text{ oe}$

A1: States in the year 2022. A candidate using a GP formula can be awarded full marks
 Allow different methods in part (c).
 Eg Via GP a formula

M1: $24800 \times 1.0658^{n-1} = 100000 \Rightarrow 1.0658^{n-1} = K$

dM1: Uses a correct method to find n .

A2: 2022

Via (trial and improvement)

M1: Uses the model by substituting integer values of t into their $V = Ap^t$ so that for $t = n, V < 100\,000$ or $t = n+1, V > 100\,000$

(So for the correct A and p this would be scored for $t = 21, V \approx £95\,000$ or $t = 21, V \approx £101\,000$)

dM1: For a complete method showing that this is the least value. So both of the above values

A1: Allow for 22 following correct and accurate results (awrt nearest £1000 is sufficient accuracy)

A1: As before

Q5.

Question	Scheme	Marks	AOs
(a)	$k = e^x$ or $x \neq e^x$	B1	2.2a
		(1)	

(b)	$g'(x) = \frac{(\ln x - 2) \times \frac{3}{x} - (3 \ln x - 7) \times \frac{1}{x}}{(\ln x - 2)^2} = \frac{1}{x(\ln x - 2)^2}$ or $g'(x) = \frac{d}{dx} \left(3 - (\ln(x) - 2)^{-1} \right) = (\ln x - 2)^{-2} \times \frac{1}{x} = \frac{1}{x(\ln x - 2)^2}$ or $g'(x) = (\ln x - 2)^{-1} \times \frac{3}{x} - (3 \ln x - 7)(\ln x - 2)^{-2} \times \frac{1}{x} = \frac{1}{x(\ln x - 2)^2}$	M1 A1	1.1b 2.1
	As $x > 0$ (or $1/x > 0$) AND $\ln x - 2$ is squared so $g'(x) > 0$	A1cso	2.4
	(3)		
(c)	Attempts to solve either $3 \ln x - 7 \dots 0$ or $\ln x - 2 \dots 0$ or $3 \ln a - 7 \dots 0$ or $\ln a - 2 \dots 0$ where $\dots$ is " $=$ " or " $>$ " to reach a value for x or a but may be seen as an inequality e.g. $x > \dots$ or $a > \dots$	M1	3.1a
	$0 < a < e^2, \quad a > e^{\frac{7}{3}}$	A1	2.2a
	(2)		
(6 marks)			

Notes:

(a)

B1: Deduces $k = e^2$ or $x \neq e^2$ Condone $k = \text{awrt } 7.39$ or $x \neq \text{awrt } 7.39$

(b)

M1: Attempts to differentiate via the quotient rule and with $\ln x \rightarrow \frac{1}{x}$ so allow for:

$$\frac{d}{dx}(g(x)) = \frac{(\ln x - 2) \times \frac{\alpha}{x} - (3 \ln x - 7) \times \frac{\beta}{x}}{(\ln x - 2)^2}, \quad \beta > 0$$

But a correct rule may be implied by their u, v, u', v' followed by applying $\frac{vu' - uv'}{v^2}$ etc.

Alternatively attempts to write $g(x) = \frac{3 \ln(x) - 7}{\ln(x) - 2} = 3 - (\ln(x) - 2)^{-1}$ and attempts the chain rule so allow for:

$$3 - (\ln(x) - 2)^{-1} \rightarrow (\ln(x) - 2)^{-2} \times \frac{\alpha}{x}$$

Alternatively writes $g(x) = (3 \ln(x) - 7)(\ln(x) - 2)^{-1}$ and attempts the product rule so allow for:

$$g'(x) = (\ln x - 2)^{-1} \times \frac{\alpha}{x} - (3 \ln x - 7)(\ln x - 2)^{-2} \times \frac{\beta}{x}$$

In general condone missing brackets for the M mark. E.g. if they quote $u = 3 \ln x - 7$ and $v = \ln x - 2$ and don't make the differentiation easier, they can be awarded this mark for applying the correct rule. Also allow this mark if they quote the correct quotient rule but only have v rather than v^2 in the denominator.

A1: $\frac{1}{x(\ln x - 2)^2}$ Allow $\frac{\frac{1}{x}}{(\ln x - 2)^2}$ i.e. we need to see the numerator simplified to $1/x$

Note that some candidates establish the correct numerator and correct denominator independently and provided they obtain the correct expressions, this mark can be awarded.

But allow a correctly expanded denominator.

A1cso: States that as $x > 0$ AND $\ln x - 2$ is squared so $g'(x) > 0$

(c)

M1: Attempts to solve either $3 \ln x - 7 = 0$ or $\ln x - 2 = 0$ or using inequalities e.g. $3 \ln x - 7 > 0$

A1: $0 < a < e^2, \quad a > e^{\frac{7}{3}}$

Q6.

Question	Scheme	Marks	AOs
(a)	$T = al^b \Rightarrow \log_{10} T = \log_{10} a + b \log_{10} l$	M1	2.1
	$\Rightarrow \log_{10} T = \log_{10} a + b \log_{10} l$ or $\Rightarrow \log_{10} T = b \log_{10} l + \log_{10} a$	A1*	1.1b
		(2)	
(b)	$b = 0.495$ or $b = \frac{45}{91}$	B1	2.2a
	$0 = "0.495" \times -0.7 + \log_{10} a \Rightarrow a = 10^{0.346...}$ or $0.45 = "0.495" \times 0.21 + \log_{10} a \Rightarrow a = 10^{0.346...}$	M1	3.1a
	$T = 2.22l^{0.495}$	A1	3.3
		(3)	
(c)	The time taken for one swing of a pendulum of length 1 m	B1	3.2a
		(1)	
(6 marks)			

Notes
(a) M1: Takes logs of both sides and shows the addition law. Implied by $T = al^b \Rightarrow \log_{10} a + b \log_{10} l$ A1*: Uses the power law to obtain the given equation with no errors. Allow the bases to be missing in the working but they must be present in the final answer. Also allow t rather than T and A rather than a. Allow working backwards e.g. $\log_{10} T = b \log_{10} l + \log_{10} a \Rightarrow \log_{10} T = \log_{10} l^b + \log_{10} a$ $\Rightarrow \log_{10} T = \log_{10} al^b \Rightarrow T = al^b$ M1: Uses the given answer and uses the power law and addition law correctly A1: Reaches the given equation with no errors as above (b) B1: Deduces the correct value for b (Allow awrt 0.495 or $\frac{45}{91}$) M1: Correct strategy to find the value of a. E.g. substitutes one of the given points and their value for b into $\log_{10} T = \log_{10} a + b \log_{10} l$ and uses correct log work to identify the value of a. Allow slips in rearranging their equation but must be correct log work to find a. Alternatively finds the equation of the straight line and equates the constant to $\log_{10} a$ and uses correct log work to identify the value of a. E.g. $y - 0.45 = "0.495"(x - 0.21) \Rightarrow y = "0.495"x + 0.346 \Rightarrow a = 10^{0.346} = ...$ A1: Complete equation $T = 2.22l^{0.495}$ or $T = 2.22l^{\frac{45}{91}}$ (Allow awrt 2.22 and awrt 0.495 or $\frac{45}{91}$) Must see the <u>equation</u> not just correct values as it is a requirement of the question. (c) B1: Correct interpretation

Q7.

Question	Scheme	Marks	AOs
(a)	$\frac{dV}{dt} = 0.48 - 0.1h$	B1	3.1b
	$V = 24h \Rightarrow \frac{dV}{dh} = 24 \text{ or } \frac{dh}{dV} = \frac{1}{24}$	B1	3.1b
	$\frac{dh}{dt} = \frac{dV}{dt} \times \frac{dh}{dV} = \frac{0.48 - 0.1h}{24}$ or e.g. $\frac{dV}{dt} = \frac{dV}{dh} \frac{dh}{dt} \Rightarrow 0.48 - 0.1h = 24 \frac{dh}{dt}$	M1	2.1
	$1200 \frac{dh}{dt} = 24 - 5h^*$	A1*	1.1b
		(4)	
(b)	$1200 \frac{dh}{dt} = 24 - 5h \Rightarrow \int \frac{1200}{24 - 5h} dh = \int dt$ $\Rightarrow \text{e.g. } \alpha \ln(24 - 5h) = t(+c) \text{ oe}$ or $1200 \frac{dh}{dt} = 24 - 5h \Rightarrow \frac{dt}{dh} = \frac{1200}{24 - 5h}$ $\Rightarrow \text{e.g. } t(+c) = \alpha \ln(24 - 5h) \text{ oe}$	M1	3.1a
	$t = -240 \ln(24 - 5h)(+c) \text{ oe}$	A1	1.1b
	$t = 0, h = 2 \Rightarrow 0 = -240 \ln(24 - 10) + c \Rightarrow c = \dots(240 \ln 14)$	M1	3.4
	$t = 240 \ln(14) - 240 \ln(24 - 5h)$	A1	1.1b
	$t = 240 \ln \frac{14}{24 - 5h} \Rightarrow \frac{t}{240} = \ln \frac{14}{24 - 5h} \Rightarrow e^{\frac{t}{240}} = \frac{14}{24 - 5h}$ $\Rightarrow 14e^{-\frac{t}{240}} = 24 - 5h \Rightarrow h = \dots$	ddM1	2.1
	$h = 4.8 - 2.8e^{-\frac{t}{240}} \text{ oe e.g. } h = \frac{24}{5} - \frac{14}{5}e^{-\frac{t}{240}}$	A1	3.3
		(6)	



(c)	Examples: - As $t \rightarrow \infty$, $e^{-\frac{t}{240}} \rightarrow 0$ - When $h > 4.8$, $\frac{dV}{dt} < 0$ - Flow in = flow out at max h so $0.1h = 4.8 \Rightarrow h = 4.8$ - As $e^{-\frac{t}{240}} > 0$, $h < 4.8$ $h = 5 \Rightarrow \frac{dV}{dt} = -0.02$ or $\frac{dh}{dt} = -\frac{1}{1200}$ $\frac{dh}{dt} = 0 \Rightarrow h = 4.8$ $h = 5 \Rightarrow 4.8 - 2.8e^{-\frac{t}{240}} = 5 \Rightarrow e^{-\frac{t}{240}} < 0$	M1	3.1b
	- The limit for h (according to the model) is 4.8m and the tank is 5m high so the tank will never become full - If $h = 5$ the tank would be emptying so can never be full - The equation can't be solved when $h = 5$	A1	3.2a
		(2)	
(12 marks)			

Notes

(a)

B1: Identifies the correct expression for $\frac{dV}{dt}$ according to the modelB1: Identifies the correct expression for $\frac{dV}{dh}$ according to the modelM1: Applies $\frac{dh}{dt} = \frac{dV}{dt} \times \frac{dh}{dV}$ or equivalent correct formula with their $\frac{dV}{dt}$ and $\frac{dV}{dh}$ which may be implied by their working

A1*: Correct equation obtained with no errors

Note that: $\frac{dV}{dt} = 0.48 - 0.1h \Rightarrow \frac{dh}{dt} = \frac{0.48 - 0.1h}{24} \Rightarrow 1200 \frac{dh}{dt} = 24 - 5h$ * scores

B1B0M0A0. There must be clear evidence where the "24" comes from and evidence of the correct chain rule being applied.



(b)

M1: Adopts a correct strategy by separating the variables correctly or rearranges to obtain $\frac{dt}{dh}$

correctly in terms of h and integrates to obtain $t = \alpha \ln(24 - 5h)(+c)$ or equivalent (condone missing brackets around the " $24 - 5h$ ") and $+c$ not required for this mark.

A1: Correct equation in any form and $+c$ not required. Do not condone missing brackets unless they are implied by subsequent work.

M1: Substitutes $t = 0$ and $h = 2$ to find their constant of integration (there must have been some attempt to integrate)

A1: Correct equation in any form

ddM1: Uses fully correct log work to obtain h in terms of t .

This depends on both previous method marks.

A1: Correct equation

Note that the marks may be earned in a different order e.g.:

$$t + c = -240 \ln(24 - 5h) \Rightarrow -\frac{t}{240} + d = \ln(24 - 5h) \Rightarrow Ae^{-\frac{t}{240}} = 24 - 5h$$

$$t = 0, h = 2 \Rightarrow A = 14 \Rightarrow 14e^{-\frac{t}{240}} = 24 - 5h \Rightarrow h = 4.8 - 2.8e^{-\frac{t}{240}}$$

Score as M1 A1 as in main scheme then

M1: Correct work leading to $Ae^{-\frac{t}{240}} = 24 - 5h$ (must have a constant "A")

$$A1: Ae^{-\frac{t}{240}} = 24 - 5h$$

ddM1: Uses $t = 0, h = 2$ in an expression of the form above to find A

$$A1: h = 4.8 - 2.8e^{-\frac{t}{240}}$$

(c)

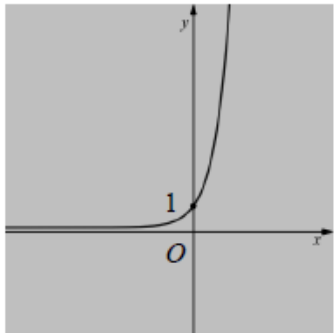
M1: See scheme for some examples

A1: Makes a correct interpretation for their method.

There must be no incorrect working or contradictory statements.

This is not a follow through mark and if their equation in (b) is used it must be correct.

Q8.

Question	Scheme	Marks	AOs
(a)		B1	1.2
	Fully correct – see notes	B1	1.1b
		(2)	
(b)	$4^x = 100 \Rightarrow x = \log_4 100$ or e.g. $x \log 4 = \log 100 \Rightarrow x = \frac{\log 100}{\log 4}$	M1	1.1b
	$\Rightarrow (x =) \text{awrt } 3.32$	A1	1.1b
		(2)	
			(4 marks)
Notes:			

Note that B0B1 is not possible in part (a)

(a) Axes do not need to be labelled. No sketch is no marks.

B1: Correct shape or correct intercept.

Shape: A positive exponential curve in quadrants 1 and 2 only, passing through a point on the positive y-axis. Must “level out” in quadrant 2 but not necessarily asymptotic to the x-axis and allow if the curve bends up slightly for $x < 0$ but do not allow a clear “U” shape. It must not clearly “stop” on the x-axis to the left of the y-axis.

OR

Intercept: The intercept can be marked as 1 or (0, 1) or $y = 1$ or (1, 0) as long as it is in the correct place. May also be seen away from the sketch but must be seen as (0, 1) or possibly these coordinates in a table but it must correspond to the sketch. If there is any ambiguity, the sketch takes precedence.

B1: Fully correct.

Shape: A positive exponential curve in quadrants 1 and 2 only, passing through a point on the positive y-axis. The curve must appear to be asymptotic to the x-axis and it must level out at least half way below the intercept. Allow if the curve bends up slightly for $x < 0$ but do not allow a clear “U” shape. The curve must not bend back on itself on the rhs of the y-axis. There must be no suggestion that the curve approaches another horizontal asymptote other than the x-axis e.g. a horizontal dotted line that the curve approaches.

AND

Intercept: As above

See practice items and below for some examples:

(b)

M1: Uses logs in an attempt to solve the equation. E.g. takes log base 4 and obtains $x = \log_4 100$

Alternatively takes logs (any base) to obtain $x \log 4 = \log 100$ and proceeds to $x = \frac{\log 100}{\log 4}$

Allow if this subsequently becomes e.g. $\log 25$ as long as $\frac{\log 100}{\log 4}$ is seen but

$x \log 4 = \log 100 \Rightarrow x = \log 25$ or $x \log 4 = \log 100 \Rightarrow x = \log 100 - \log 4$ scores M0

A1: awrt 3.32 . A correct answer only of awrt 3.32 scores M1A1

Note that a common incorrect answer is $x = 3.218875\dots$ and comes from $\ln 25$ or $\ln 100 - \ln 4$

and unless $x = \frac{\ln 100}{\ln 4}$ is seen previously, this scores M0A0

Q9.

Part	Working or answer an examiner might expect to see	Mark	Notes
(a)	If $d = kV^n$, then $\log_{10} d = \log_{10} k + n \log_{10} V$	M1	This mark is given for
	Plotting $\log_{10} d$ against $\log_{10} V$ will result in a straight line with gradient n and intercept $\log_{10} k$	A1	This mark is given for an explanation of why the second graph shows that $d = kV^n$
	$\log_{10} k = -1.77$ $k = 10^{-1.77} = 0.01698\dots \approx 0.017$	A1	This mark is for showing fully that $k \approx 0.017$
(b)	$d = kV^n$ When $V = 30$, $d = 20$ and $k = 0.17$ then $20 = 0.017 \times 30^n$	M1	This mark is given for substituting in the formula as a method to find the value of n
	$n \log 30 = \log \left(\frac{20}{0.017} \right)$	M1	This mark is given for a correct expression for n
	$n = 2.08$ to 3 significant figures $d = 0.017 \times V^{2.08}$	A1	This mark is given for finding a correct value of n to 3 significant figures and writing a complete equation for the model
(c)	$\frac{60}{3600} \times 0.8 \times 1000 = 13.33 \text{ m}$	M1	This mark is given for a method to find the distance, in metres, covered in the reaction time of 0.8 seconds
	$d = 0.017 \times 60^{2.08} = 84.92 \text{ m}$	M1	This mark is given for a method to use the formula to find the stopping distance
	$13.33 \text{ m} + 84.92 \text{ m} = 98.25 \text{ m}$ Sean will be able to stop before reaching the puddle	A1	This mark is given for finding a correct value of the total stopping distance and giving a valid conclusion
(Total 9 marks)			

Q10.

Question	Scheme	Marks	AOs
(a)	265 thousand	B1	3.4
		(1)	
(b)	Attempts $\frac{dN_b}{dt} = 11e^{0.05t}$	M1	1.1b
	Substitutes $t=10$ into their $\frac{dN_b}{dt}$	M1	3.4
	$\frac{dN_b}{dt} = \text{awrt } 18.1$ which is approximately 18 thousand per year *	A1*	2.1
		(3)	
(c)	Sets $45 + 220e^{0.05t} = 10 + 800e^{-0.05t} \Rightarrow 220e^{0.05t} + 35 - 800e^{-0.05t} = 0$	M1	3.1b
	Correct quadratic equation $\Rightarrow 220(e^{0.05t})^2 + 35e^{0.05t} - 800 = 0$	A1	1.1b
	$e^{0.05t} = 1.829, (-1.988) \Rightarrow 0.05t = \ln(1.829)$	M1	2.1
	$T = 12.08$	A1	1.1b
		(4)	
(8 marks)			
Notes:			

(a) May be seen in the question so watch out.

B1: Accept 265 thousand or 265 000 or equivalent such as 265 k but not just 265.

(b)

M1: Differentiates to a form $ke^{0.05t}$, $k > 0$, $k \neq 220$. Do not be too concerned about the lhs.

M1: Substitutes $t=10$ into a changed function that was formed from an attempt at differentiation.

The left hand side must have implied differentiation. E.g. Rate = , N' , $\frac{dN_b}{dt}$, $\frac{dN}{dt}$ or even $\frac{dy}{dx}$

A1*: Full and complete proof that requires

- some correct lhs seen at some point. E.g. "Rate = , " $\frac{dN_b}{dt}$, $\frac{dN}{dt}$ but condone N' .
- an intermediate line/answer of either $11e^{0.05 \times 10}$ or awrt 18.1 before a minimal conclusion which must be referencing the 18 000 or 18 thousand



(c)

M1: Attempts to set both equations equal to each other and simplify the constant terms.

Look for $220e^{0.05t} + 35 = 800e^{-0.05t}$ o.e but condone slipsIt is also possible to set $\frac{N-45}{220} = \left(e^{0.05t}\right) = \frac{800}{N-10}$ and form an equation in N

A1: Correct quadratic form.

Look for $220(e^{0.05t})^2 + 35e^{0.05t} - 800 = 0$ or $220e^{0.1t} + 35e^{0.05t} - 800 = 0$ but allow with terms in different order such as $220e^{0.1t} + 35e^{0.05t} = 800$ FYI the equation in N is $N^2 - 55N - 175550 = 0$ M1: Full attempt to find the value of t (or a constant multiple of t)This involves the key step of recognising and solving a 3TQ in $e^{0.05t}$ followed by the use of lns.

If the answers to the quadratic just appear (from a calculator) you will need to check.

Accuracy should be to 3sf.

You may see different variables used such as x

$$x = e^{0.05t}, 220e^{0.1t} + 35e^{0.05t} = 800 \Rightarrow 220x^2 + 35x = 800 \Rightarrow x = 1.82... \Rightarrow t = 20 \ln 1.82...$$

Allow use of calculator for solving the quadratic and for $e^{0.05t} = 1.82... \Rightarrow t = 12.08$ Via the N route it will involve substituting the positive solution to their quadratic into either equation to find a value for t/T using same rules as above.

A1: AWRT 12.08

Answers with limited or no working in (b) and (c)

(b) A derivative in the correct form must be seen

(c) Candidates who state $45 + 220e^{0.05t} = 10 + 800e^{-0.05t}$ followed by awrt 12.08 (presumably from using num-solv on their calculators) can score SC 1100. Rubric on the front of the paper states that "Answers without working may not gain full credit" so we demand a method in this part.